LKT augmented GRU with a 128-unit prediction head

Saved model: augmented__head128__seed1 | restored epoch 24 | 31,585 parameter entries

A Inputs available before response t

5 numeric coordinates (in saved order)
log(1 + student time gap in seconds)
intercept; student-item logitdec;
student-item PPES; student logitdec
Standardize using training-set statistics

Current item ID: KC..Default.
144 known items

Learned item embedding: 16 values
Table: 145 rows × 16 columns
144 known rows + 1 reserved zero row

Current skill ID: KC..Cluster.
36 known skills

Learned skill embedding: 16 values
Table: 37 rows × 16 columns
36 known rows + 1 reserved zero row

Concatenate: 5 numeric + 16 item + 16 skill = 37 coordinates
Input dropout 0.1 produces x(t), shared by prediction and memory update

Embeddings are trained jointly with the network. The centered intercept coordinate is zero; no problem-text embedding is used.

B Predict the current response

Previous top-layer memory

Previous h3(t−1): 32
+ current x(t): 37
69 coordinates

Linear 69 → 128 (with bias)
ReLU
Head dropout 0.1

Linear 128 → 1
(with bias)
One logit

Sigmoid + clipping
p(correct at trial t)
ε = 10^{-5}

Current response y(t) does not enter this prediction. No separate inference network or Gaussian latent is present.

C Observe the answer, then update all three GRU layers

Observed response y(t)
0 = incorrect; 1 = correct

h1(t−1)
32 previous values

h2(t−1)
32 previous values

h3(t−1)
32 previous values

x(t): 37

Concatenate [x(t), y(t)]
37 + 1 = 38 coordinates

GRU layer 1
32 units
Own state

h1(t): 32

Dropout
0.1

GRU layer 2
32 units
Own state

h2(t): 32

Dropout
0.1

GRU layer 3
32 units
Own state

h3(t)
32 values

Retain all three states for the next trial: [h1(t), h2(t), h3(t)] = 3 × 32 values

Each new participant starts with zero states. Updated memory can affect the NEXT prediction, p(t+1).

Dropout is active only during training. Binary cross-entropy trains the logits; all learned weights stay fixed during continuation.